Prediction of AKI in ICU using MIMIC-IV: From Phenotyping to Drug Attribution

Presented by **TEAM 7**

A comprehensive clinical research project investigating adverse drug events and acute kidney injury in critically ill patients using machine learning and electronic health record data.





The Challenge of Drug-Induced AKI in the ICU

Reproducible Pipelines

Many published models rely on ad-hoc preprocessing that is difficult to reproduce across institutions and datasets.

Outcome Definitions

AKI definitions vary (SCr-only vs full KDIGO), and model features often ignore temporal context and progression.

Class Imbalance

AKI events attributed to specific drugs are rare, so metrics like ROC-AUC can look impressive even when precision and F1 are poor.

Drug-induced acute kidney injury: a cohort study on incidence, identification of pathophysiological mechanisms, and prognostic factors

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Introduction: Acut
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Methods: A retrosp
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Clinical Kidney Journal

Clin Kidney J. 2022 Aug 2;15(12):2266–2280. doi: 10.1093/ckj/sfac181

Machine learning models for predicting acute kidney injury: a systematic review and critical appraisal

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Background

The number of studies applying machine learning (ML) to predict acute kidney injury (AKI) has grown steadily over the past decade. We assess and critically appraise the state of the art in ML models for AKI prediction, considering performance, methodological soundness, and applicability.

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EDITORIAL

Reproducibility in Research: The Role of *Kidney360*

Allon, Michael[1]; Juncos, Luis A.[2]; Perazella, Mark A.[3]

Author Information

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OPEN

Reproducibility is one of the hallmarks of scientific research. Independent confirmation of an original research paper increases confidence in the validity of its findings. Conversely, publication of a study that contradicts the findings of the first paper raises doubt about the conclusions of the original paper. As the editors of *Kidney360*, we think it is critical for us to publish both confirmatory studies and negative studies to counterbalance positive studies that have been published previously. Moreover, we struggle to ensure that studies published by pharmaceutical companies are as free of conflict of interest as possible.

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[Submitted on 13 Aug 2025]

Interpretable Machine Learning Model for Early Prediction of Acute Kidney Injury in Critically Ill Patients with Cirrhosis: A Retrospective Study

Li Sun, Shuheng Chen, Junyi Fan, Yong Si, Minoo Ahmadi, Elham Pishgar, Kamiar Alaei, Maryam Pishgar

Background: Cirrhosis is a progressive liver disease with high mortality and frequent complications, notably acute kidney injury (AKI), which occurs in up to 50% of hospitalized patients and worsens outcomes. AKI stems from complex hemodynamic, inflammatory, and metabolic changes, making early detection essential. Many
acy, interpretability, and alignment with intensive care unit (ICU) workflows. This study developed an interpretable machine learning model for
ically ill patients with cirrhosis.
retrospective analysis of the MIMIC-IV v2.2 databa
key data. Laboratory and physiological variables f
lection, and SMOTE class balancing. Six algorithms
AUROC, accuracy, F1-score, sensitivity, specificit
d the best performance (AUROC 0.808, 95% CI 0.7
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Machine learning for risk prediction of acute kidney injury in patients with diabetes mellitus combined with heart failure during hospitalization

Guojing Li, Zhiqiang Zhao, Zongliang Yu, Junyi Liao & Mengyao Zhang

Scientific Reports 15, Article number: 10728 (2025) | Cite this article

4499 Accesses | 2 Citations | 1 Altmetric | Metrics

Abstract

This study aimed to develop a machine learning (ML) model for predicting the risk of acute kidney injury (AKI) in diabetic patients with heart failure (HF) during hospitalization. Using data from 1,457 patients in the MIMIC-IV database, the study identified twenty independent risk factors for AKI through LASSO regression and logistic regression. Six ML algorithms were evaluated, including LightGBM, random forest, and neural networks. The LightGBM model demonstrated superior performance with the highest prediction accuracy, with AUC values of

Clinical & Technical Motivation

AKI is common, deadly, and costly in ICU patients

Many AKI cases are drug-associated but preventable

Clinicians detect AKI late—after kidney damage is already done

ICU data is too complex to track manually (labs, vitals, UO, drug timing)

Existing AKI models ignore drug timing and are not reproducible



Who Benefits From This

ICU pharmacists

Dose adjustment, nephrotoxin stewardship

Nephrologists

Early detection of at-risk patients

ICU physicians

Safer medication decision-making

ICU nurses

Closer monitoring for flagged patients

Quality & safety teams

Track preventable drug-related harm

Clinical informatics teams

Need reproducible SQL pipelines

Patients

Ultimate beneficiaries of reduced AKI harm

Impact of Our Model

Early Intervention

Pharmacists act before harm

Escalated Monitoring

Nephrologists adjust therapy sooner

Real-Time Awareness

Physicians see drug risk instantly

Tighter Tracking

Nurses monitor vitals & urine output

Stewardship Support

Hospital nephrotoxin programs

Reduced Harm

Less AKI, dialysis, ICU days

Reproducible Pipeline

Transparent, adoptable SQL

Key Considerations

Temporal association ≠ causality; AKI after drug start treated as possibly drug-related

48-hour window assumed reasonable for drug-induced AKI onset

Baseline creatinine (lowest in 7 days) assumed to represent true kidney function

KDIGO-based onset time inferred from labs/UO assumed sufficiently accurate

Confounders (sepsis, vasopressors, ventilation) only partially captured

High recall with modest precision assumed acceptable for screening workflows

Our Approach: A Four-Part Solution

1

SQL-Defined ETL Pipeline

Built a fully SQL-based BigQuery pipeline on MIMIC-IV v3.1 to generate KDIGO AKI labels and drug-level ADE outcomes.

2

Supervised Learning Tasks

Designed two prediction tasks—48-hour drug-associated AKI and a broader “any post-drug AKI” outcome.

3

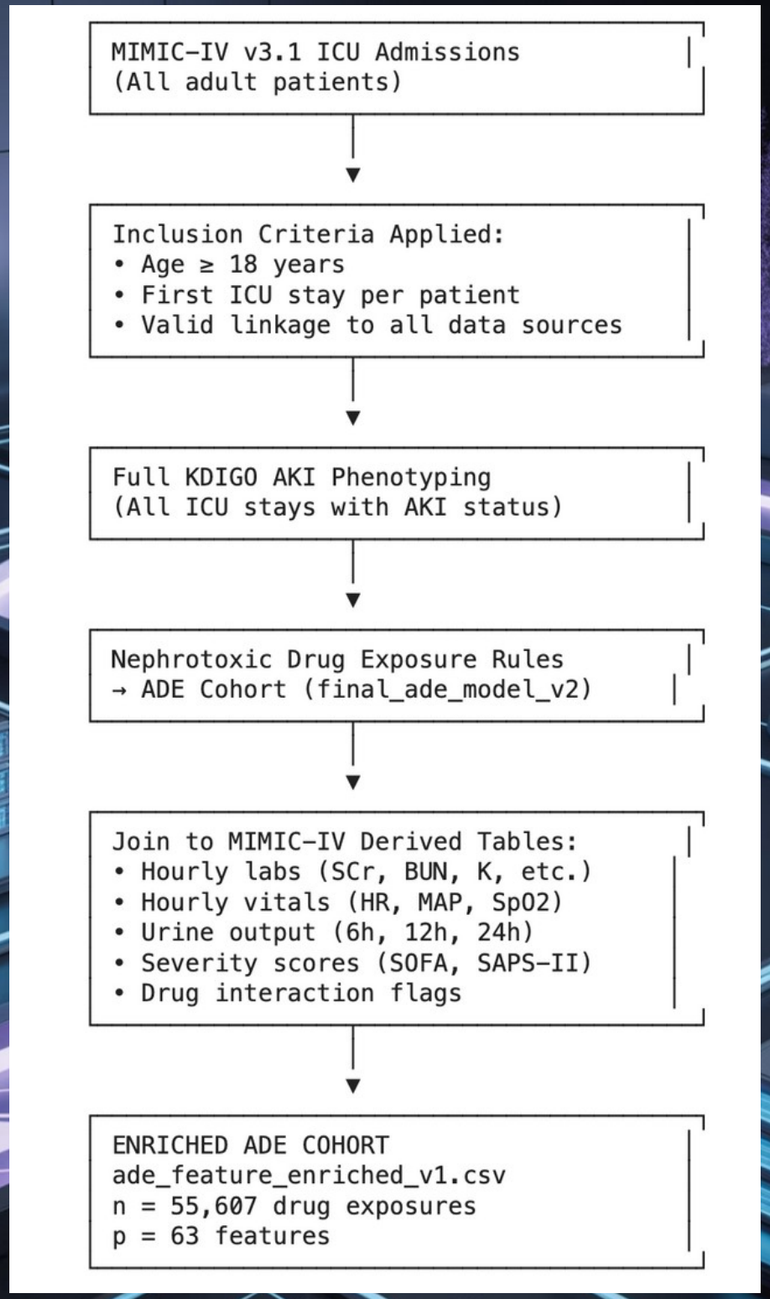
Feature Comparison

Tested baseline vs. temporally enriched models using rolling labs, vitals, urine output, severity scores, and key clinical interactions.

4

Clinical Interpretation

Assessed models with PR-AUC and threshold metrics, and used SHAP for clinically meaningful explanations.



Data Source and Cohort Construction

MIMIC-IV v3.1 on BigQuery

All analysis used MIMIC-IV v3.1 on Google BigQuery, pulling from core hospital tables and ICU tables..

All transformations were implemented using SQL views/CREATE OR REPLACE TABLE for fully reproducible pipelines.

Adult ICU Cohort

We constructed an adult ICU cohort including all ICU stays for patients >=18 years at ICU admission. For patients with multiple stays, we kept only the first ICU stay

Final cohort: 65,366 unique ICU stays

Table 1. Baseline Characteristics of the Enriched ADE Cohort

Variable	Overall (N=55,607)	AKI (N=1,734)	No AKI (N=53,873)
Age (years)	64.0 ± 15.8	67.3 ± 15.0	63.9 ± 15.8
Sex: Male	32,725 (58.9%)	900 (51.9%)	31,825 (59.1%)
Sex: Female	22,882 (41.1%)	834 (48.1%)	22,048 (40.9%)
Race: BLACK/AFRICAN AMERICAN	4,057 (7.3%)	107 (6.2%)	3,950 (7.3%)
Race: OTHER	1,948 (3.5%)	60 (3.5%)	1,888 (3.5%)
Race: UNABLE TO OBTAIN	1,398 (2.5%)	52 (3.0%)	1,346 (2.5%)
Race: UNKNOWN	6,327 (11.4%)	248 (14.3%)	6,079 (11.3%)
Race: WHITE	34,637 (62.3%)	1,054 (60.8%)	33,583 (62.3%)
Race: WHITE - OTHER EUROPEAN	1,259 (2.3%)	46 (2.7%)	1,213 (2.3%)
Race: Other	5,981 (10.8%)	167 (9.6%)	5,814 (10.8%)
ICU Type: MICU	10,065 (18.1%)	544 (31.4%)	9,521 (17.7%)
ICU Type: SICU	14,886 (26.8%)	500 (28.8%)	14,386 (26.7%)
ICU Type: CCU	21,476 (38.6%)	535 (30.9%)	20,941 (38.9%)
ICU Type: TSICU	5,397 (9.7%)	125 (7.2%)	5,272 (9.8%)
ICU Type: Neuro ICU	3,614 (6.5%)	27 (1.6%)	3,587 (6.7%)
ICU Type: Others	169 (0.3%)	3 (0.2%)	166 (0.3%)
Chronic Kidney Disease	10,096 (18.2%)	566 (32.6%)	9,530 (17.7%)
Diabetes	18,171 (32.7%)	641 (37.0%)	17,530 (32.5%)
Congestive Heart Failure	16,032 (28.8%)	735 (42.4%)	15,297 (28.4%)
Chronic Liver Disease	5,672 (10.2%)	419 (24.2%)	5,253 (9.8%)
Cancer	7,525 (13.5%)	256 (14.8%)	7,269 (13.5%)
Charlson Index, median [IQR]	4 [3–7]	6 [4–8]	4 [3–7]
Baseline SCr (mg/dL)	1.2 ± 1.3	1.7 ± 1.6	1.2 ± 1.3
Missing	485 (0.9%)	0 (0.0%)	485 (0.9%)
Sepsis	25,563 (46.0%)	1,222 (70.5%)	24,341 (45.2%)
Vasopressor Use	22,727 (40.9%)	1,191 (68.7%)	21,536 (40.0%)
Mechanical Ventilation	47,062 (84.6%)	1,687 (97.3%)	45,375 (84.2%)
Rhabdomyolysis	1,070 (1.9%)	88 (5.1%)	982 (1.8%)
Loop Diuretics	31,626 (56.9%)	1,273 (73.4%)	30,353 (56.3%)
Drug Class: Vanco	25,384 (45.6%)	1,385 (79.9%)	23,999 (44.5%)
Drug Class: Aminoglycoside	1,726 (3.1%)	43 (2.5%)	1,683 (3.1%)
Drug Class: NSAID	9,742 (17.5%)	87 (5.0%)	9,655 (17.9%)
Drug Class: ACE_ARB	18,755 (33.7%)	219 (12.6%)	18,536 (34.4%)
Mean SCr 24h (mg/dL)	1.3 ± 1.3	1.7 ± 1.4	1.3 ± 1.3
Mean BUN 24h (mg/dL)	24.0 ± 19.1	32.0 ± 22.8	23.8 ± 18.9
Urine Output 24h (mL)	1646.6 ± 1786.7	940.6 ± 1145.5	1668.2 ± 1798.3
Drug Duration (hours)	105.7 ± 154.7	147.8 ± 193.2	104.3 ± 153.1



Defining AKI: Full KDIGO 2012 Criteria

We implemented comprehensive AKI detection using both serum creatinine (SCr) and urine output (UO) criteria, following KDIGO 2012 guidelines.

Baseline SCr

Lowest SCr in 7 days prior to ICU admission, or first SCr during hospitalization if unavailable.

SCr-Based AKI

Increase ≥0.3 mg/dL within 48 hours, ≥1.5× baseline, ≥2× baseline, or absolute SCr ≥4 mg/dL.

UO-Based AKI

Urine output <0.5 mL/kg/hr for ≥6 or ≥12 hours, <0.3 mL/kg/hr for ≥24 hours, or anuria for ≥12 hours.

Nephrotoxic Drug Exposures

We focused on five nephrotoxic drug classes, using prescriptions and ICU timing to define exposures spanning ICU stay ±1 day.

25,384

Vancomycin

Most common nephrotoxic antibiotic exposure

27,897

Loop Diuretics

Highest overall exposure count

18,755

ACE/ARBs

Common cardiovascular medications

9,742

NSAIDs

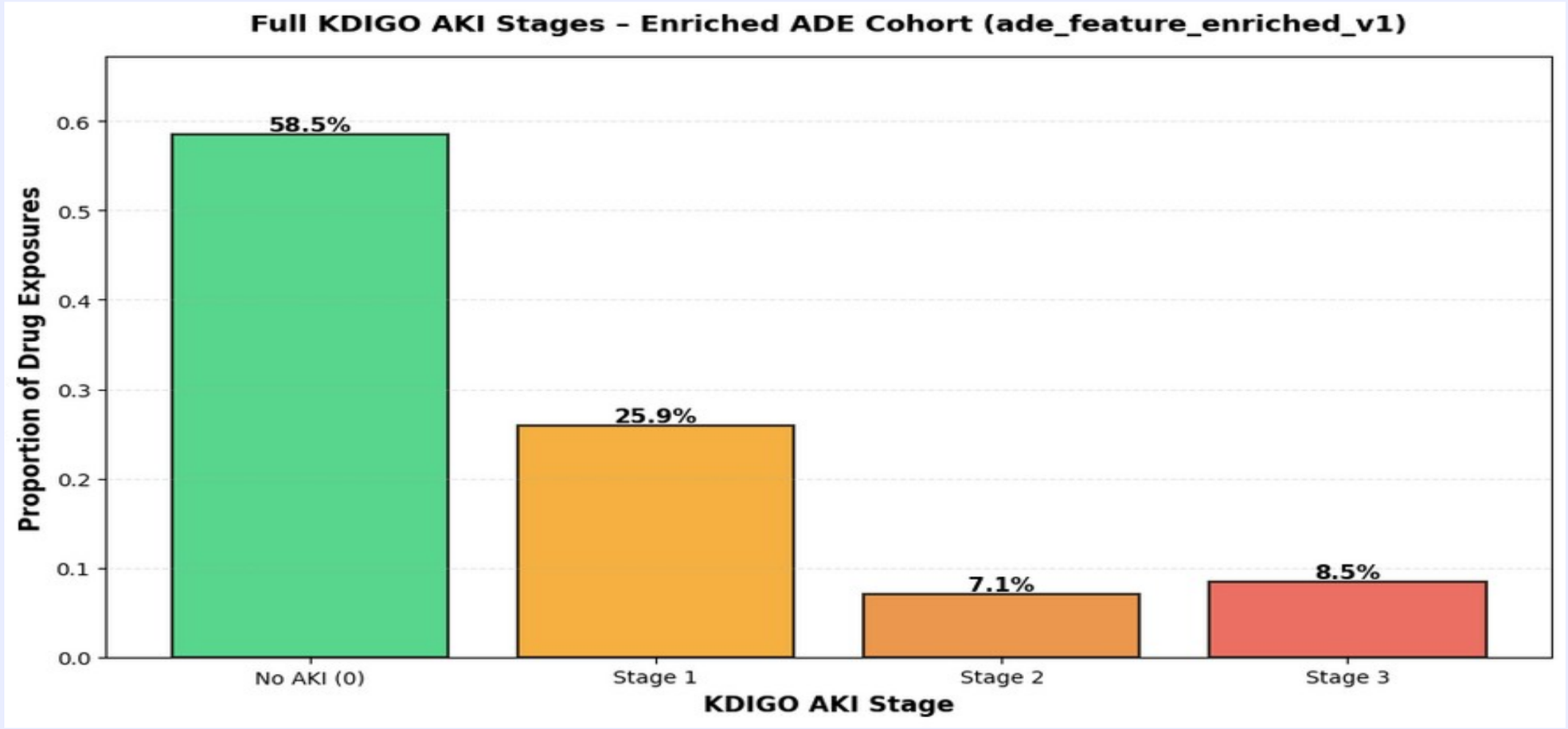
Frequently used analgesics

1,726

Aminoglycosides

Potent but nephrotoxic antibiotics

AKI Prevalence in Nephrotoxic Drugs-Exposed Cohort



Confounder Flags, ADE Outcomes, and Baseline Features

Clinical Confounder Flags

- Sepsis (Sepsis-3 criteria)
- Vasopressor use
- Mechanical ventilation
- Rhabdomyolysis (CK ≥ 5000)
- Loop diuretic exposure

Note: Independently increase AKI risk

ADE Outcomes

- 48h ADE (ade_aki48_full): KDIGO AKI within 48h post-drug. Prevalence: 3.12%
- Any post-drug AKI (any_aki_post_drug): KDIGO AKI after drug start. Prevalence: 4.20%

Baseline Features

Demographics: Age, Sex, Race, Admission type, ICU type

Kidney: Baseline SCr, SCr missing flag, CKD status

Comorbidities: Diabetes, Heart failure, Liver disease, Malignancy, Charlson index

Drug: Class (vancomycin, aminoglycoside, NSAID, ACE/ARB, loop diuretic), Start/end time

Feature Engineering: Baseline vs Enriched

Baseline Features

- Demographics & context (age, sex, race, admisssion type, ICU unit)
- Baseline kidney status(SCr, CKD indicator)
- Comorbidities (diabetes, HF, liver disease, cancer, Charlson index)
- Cofounders at drug start(sepsis, vasopressors, ventilation, rhabdomyolysis, loop exposure)
- Drug class + timing

Temporally Enriched Features

- Rolling labs (24h pre-drug): SCr, BUN, electrolytes, bicarbonate
- ICU vitals: HR, MAP, SpO2
- Urine output: 6h, 12h, 24h volumes
- Drug duration
- Severity scores(SOFA, SAPS II)
- Interaction features (e.g., CKD+vancomycin, diabetes+NSAID)



Model Training Approach

For each outcome, we trained two CatBoost classifiers using the same train/test split (80/20) with stratification by outcome.

1

Class Weighting

Applied inverse prevalence weights to address severe imbalance (31:1 for 48h AKI, 23:1 for any AKI).

2

Hyperparameters

Logistic loss, depth 6, 2000 max iterations with early stopping after 50 rounds of non-improving validation AUC.

3

Threshold Selection

Swept thresholds 0.05-0.50, identified F1-maximizing threshold, fixed at 0.50 for primary reporting.

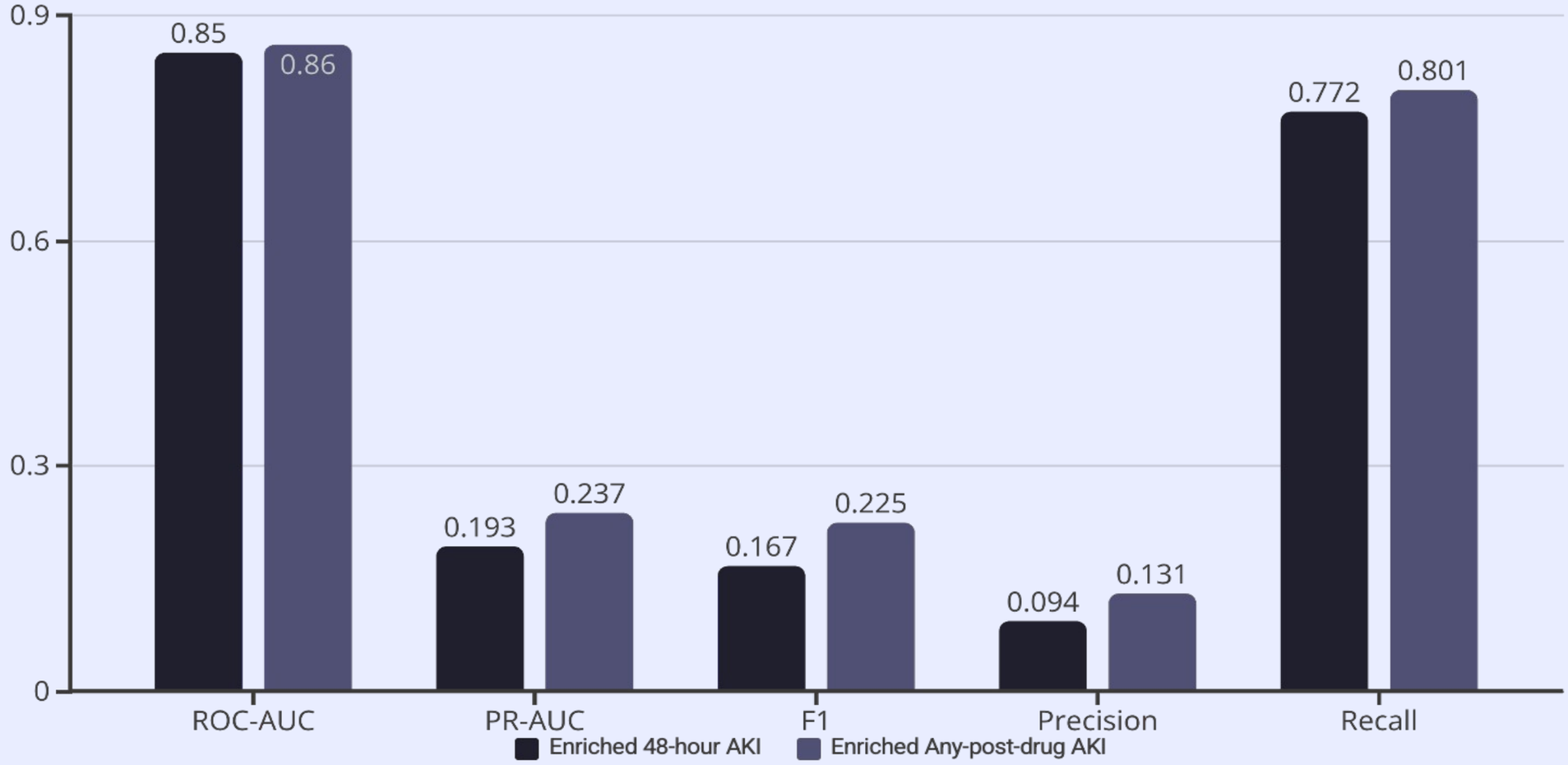
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Interpretation

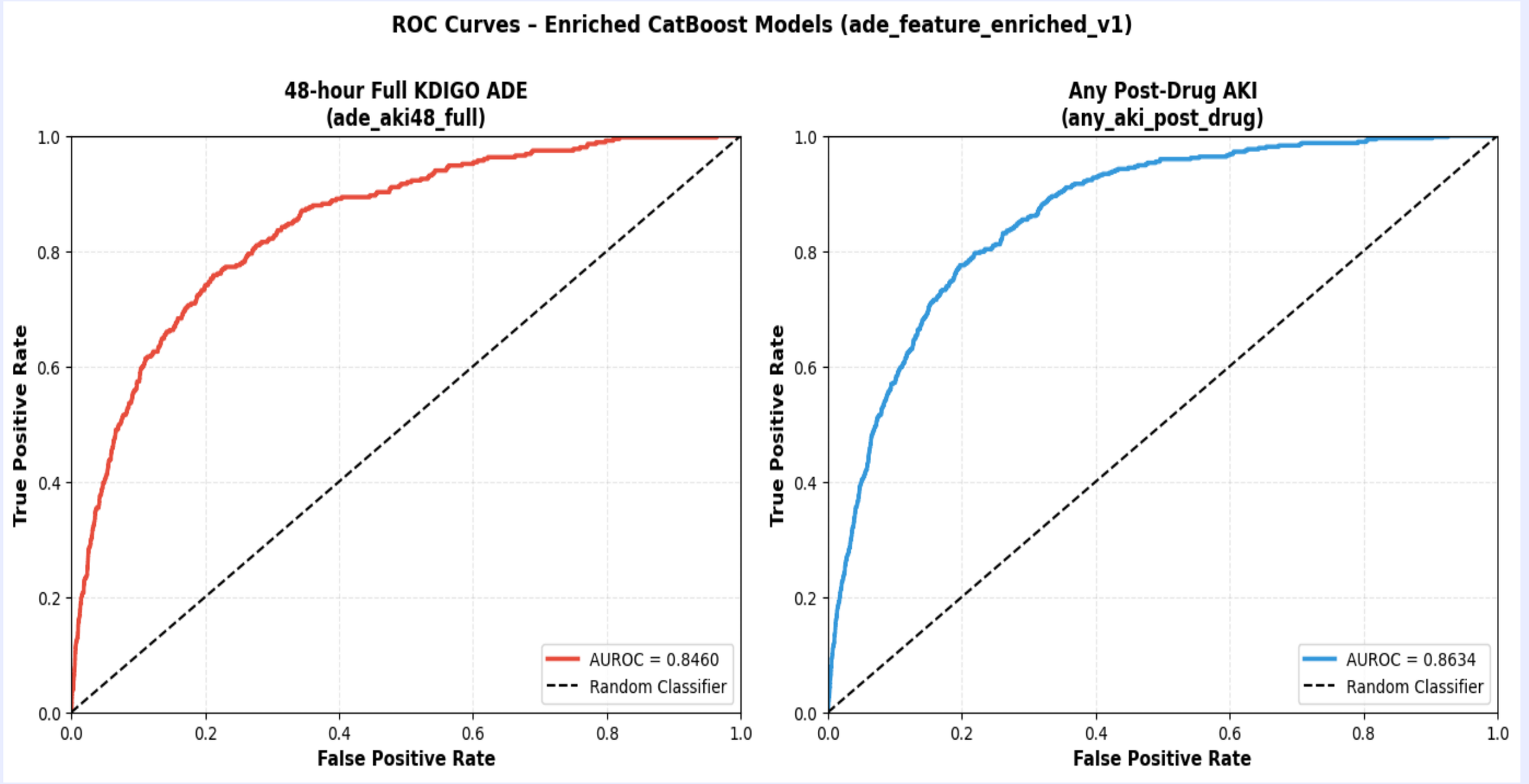
Computed SHAP values on test data to quantify feature contributions and align with clinical intuition.

Comparison of Enriched AKI Prediction Models

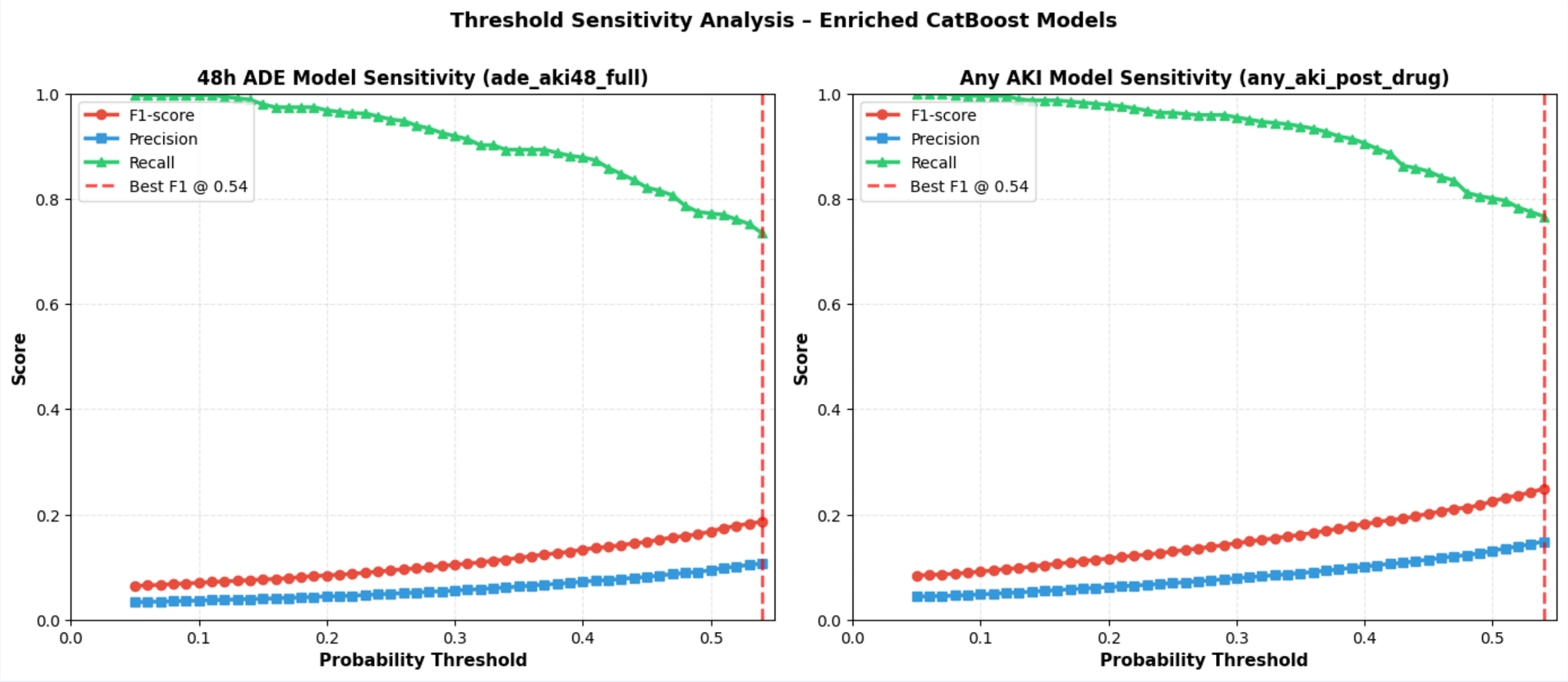
48h AKI vs Any Post-Drug AKI



The enriched Any-AKI model outperforms the enriched 48h model across PR-AUC, F1, precision, and recall. Both achieve strong ROC-AUC (0.85–0.86), but the Any-AKI model is better suited for clinical deployment.

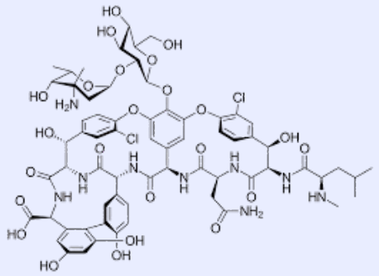

ROC Curves – Enriched CatBoost Models (ade_feature_enriched_v1)
48-hour Full KDIGO ADE
(ade_aki48_full)
True Positive Rate
False Positive Rate
AUROC = 0.8460
Random Classifier
Any Post-Drug AKI
(any_aki_post_drug)
True Positive Rate
False Positive Rate
AUROC = 0.8634
Random Classifier

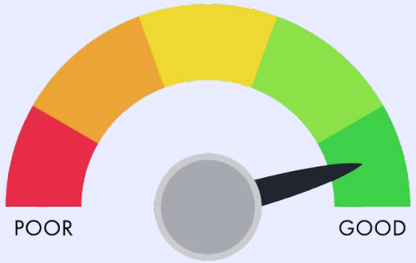
Threshold Sensitivity Analysis



What Drives AKI Risk? SHAP Analysis

SHAP analysis revealed the most important features for predicting drug-associated AKI, aligning with clinical expectations.









Vancomycin Exposure

- Strongest predictor of 48-hour AKI.
- SHAP shows high positive contribution → risk sharply increases when vancomycin is used.
- Reflects known nephrotoxicity and dose-related toxicity in critically ill patients.

SAPS II Severity Score

- Higher severity scores greatly increase AKI risk.
- Captures global physiologic instability → sicker patients are more vulnerable to kidney injury.
- SHAP confirms a *monotonic risk gradient* with rising severity.

Loop Diuretic Exposure

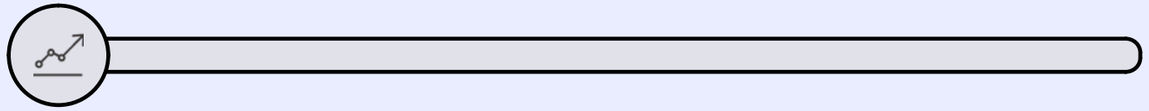
- More loop diuretic orders → fluid depletion + hemodynamic stress → increased AKI risk.
- SHAP indicates even modest diuretic exposure contributes meaningfully to model predictions.

Baseline Serum Creatinine

- Elevated baseline SCr = reduced kidney reserve.
- SHAP shows higher SCr consistently pushes prediction toward AKI.
- Clinically consistent: chronic impairment = higher susceptibility.

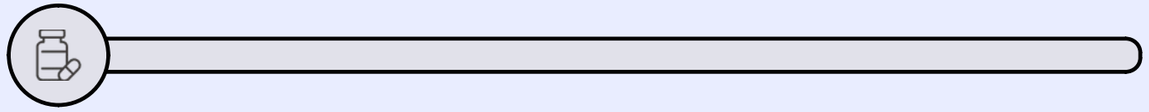
Why Temporal Enrichment Matters

Static baseline features are not enough. The largest gains came from features that capture dynamic clinical trajectories and encode the signals clinicians reason about informally.



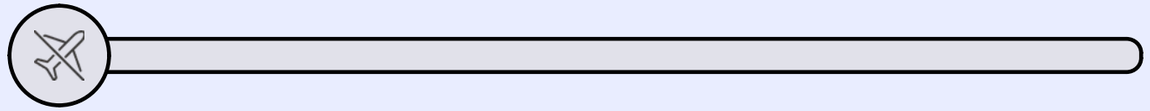
Trajectory of Kidney Function

SCr trends and low urine output revealed declining renal function before AKI onset.



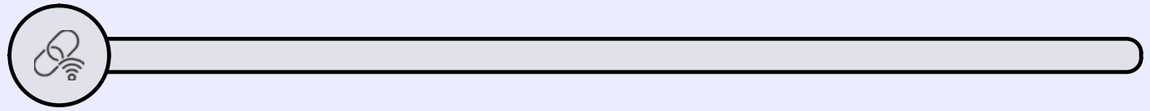
Intensity of Exposure

Loop diuretic order counts and drug duration quantified cumulative nephrotoxic burden.



Overall Severity

SOFA and SAPS II scores captured organ dysfunction and illness acuity comprehensively.



Clinical Interactions

Diabetes+NSAID and CKD+vancomycin flags marked high-risk combinations explicitly.

Encoding these signals explicitly allowed CatBoost to better separate high- and low-risk exposures, especially under severe class imbalance.

Clinical Application: Workflow

The any-AKI-post-drug model is better suited for deployment than the 48-hour model and supports a realistic screening workflow.

1

Daily Screening

Run the model once per day for patients receiving nephrotoxic drugs in the ICU.

2

Risk Flagging

At threshold 0.5, identify a subset of patients at elevated risk (recall 0.80, precision 0.13).

3

Clinical Review

Pharmacists or nephrologists review flagged cases, considering dose adjustment or alternative agents.

4

Enhanced Monitoring

Implement closer monitoring with more frequent SCr checks or urine output tracking.

Limitations and Future Directions

Current Limitations

Single-center dataset: MIMIC-IV reflects one academic medical center; external validation needed

Temporal association ≠ causation: Some events may be due to underlying illness rather than drugs

Residual confounding: Unmeasured confounders remain despite extensive adjustment

Modest precision: ~0.13 at reasonable recall limits suitability for hard-stop alerts

Future Work

- External validation on other EHR datasets and institutions
- Model calibration and monitoring for drift over time
- Integration with clinician-facing interfaces that surface SHAP-highlighted features

Conclusion: Toward Safer Medication Use in the ICU

We built and documented a **BigQuery pipeline on MIMIC-IV v3.1** that generates full KDIGO AKI labels and drug-level ADE outcomes. Temporally enriched CatBoost models substantially improved PR-AUC and recall for predicting drug-associated AKI.

Temporal Context is Crucial

Pre-drug rolling labs, vitals, urine output, severity scores, and interaction flags were essential for identifying at-risk patients.

Decision Support, Not Replacement

Models should support clinician judgment by surfacing risk scores and interpretable drivers, not make autonomous decisions.

THANK YOU!

Questions?

